Question:

Continuous Integration with Jenkins:

Tool: Jenkins

Program:

- Install and configure Jenkins.
- Create a simple pipeline to:
  - Clone a Git repository.
  - Build a sample project (e.g., a Java Maven or Python project).

Run test cases and archive the results.

Procedure:

Eclipse (Maven Project + TestNG)

↓

Push to Git Repository

↓

Jenkins Pipeline

↓

Build + Test + Report

1. Navigate to folder where maven project(1st pgm) ispresent
   C:\Users\User\eclipse-workspace\first
2. Open git bash, initialize git repo,
   - ***git init***
   - ***Open Credential manager and in windows credentials, delete git user name and email(if it is different user)***
   - ***Unset config in git bash***
     - *git config --global --unset user.name*
     - *git config --local --unset user.email*

   Set config

     - *git config --global user.name "Your Name"*
     - *git config --global user.email "you@example.com"*

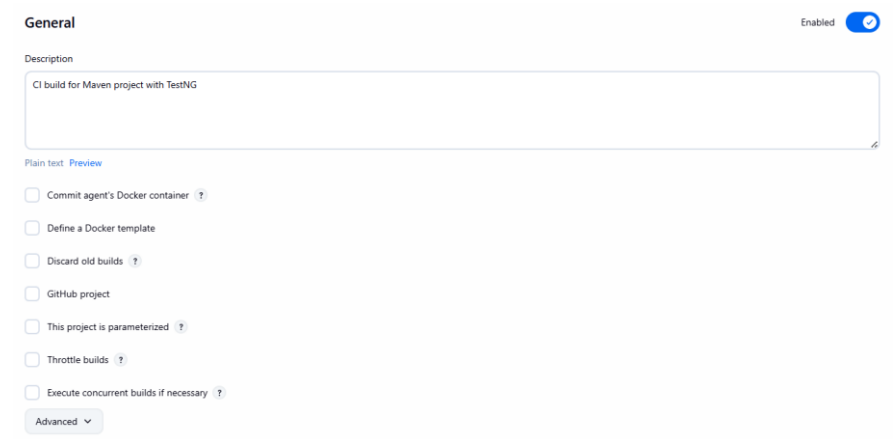
   add and commit src/ and pom
     - *git add src/ ; git commit –m "src commited"*
     - *git add pom.xml; git commit –m "pom committed"*
3. Open git hub, login to account, create new repo dev_second
4. Come back to git bash

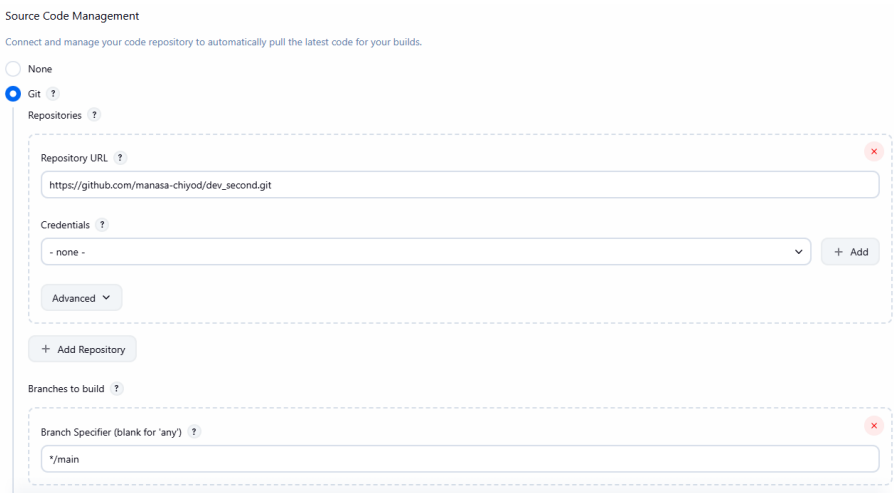
git branch -M main (change name of branch from "master" to "main")
git remote add origin <your-repo-url>
git push -u origin main

5. Jenkins Installation (Windows) - Install JDK 11 or JDK 17 (where java, remove oracle path)
6. Open browser → https://www.jenkins.io/download/
7. Download **Windows Installer (.msi)**
8. Install → keep **default options (don't use username and pwd)**
9. Jenkins runs on: http://localhost:8080
10. Enter "Initialadminpassword" location of password mentioned in same page.
11. Restart Jenkins(services.msc) and run localhost:8080
12. Go to manage Jenkins-> plugins and Install git, maven, pipeline, junit, testNG plugins.
13. Go to manage Jenkins-> tools -> set JDK and Maven names (JDK17, maven 3.9.9)
14. Create Jenkins freestyle job, configure Git, prebuild and post build steps.
15. Build the job and generate testNG report.

Output:

1. maven-freestyle-ci





Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Invoke top-level Maven targets

Maven Version

maven 3.9.9

Goals

clean test

Advanced

+ Add build step

Post-build Actions

Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.

Publish TestNG Results

TestNG XML report pattern

**/testng-results.xml

Advanced

+ Add post-build action

Jenkins / maven-freestyle-ci

Status
Changes
Workspace
Build Now
Configure
Delete Project
Rename

Builds

Filter

January 7, 2026

#1 1:31 PM

maven-freestyle-ci

Edit description

CI build for Maven project with TestNG

Latest Test Result (no failures)

Permalinks

- Last build (#1), 18 days ago
- Last stable build (#1), 18 days ago
- Last successful build (#1), 18 days ago
- Last completed build (#1), 18 days ago

Test Result Trend

Passed Skipped Failed

3
2.71
2
1.5
#1

#1
Passed 3
Skipped 0
Failed 0